LLMs Leak Secrets Through Unrelated Outputs

Anonymous Author(s)
Affiliation
Address
email

Abstract

A language model need not repeat a secret to leak it. We place a four-digit PIN in Qwen3-8B's system prompt, ask for unrelated number lists, and rank candidate PINs by the observed outputs' teacher-forced likelihood. From one list, five-way detection reaches 77.2% (chance: 20%) and remains above 73% after removing lists that contain the PIN, a digit permutation, or a Hamming-distance-1 neighbor. With approximately 20 strictly filtered lists, the attack recovers 24 of 50 held-out PINs from all 10,000 possibilities (chance: 0.01%); 36 of 50 appear in the top five. In same-model tests, seven-way detection reaches 68.6% for Qwen3-8B, 59.1% for Phi-3.5-Mini, and 44.3% for OLMo-2-7B (chance: 14.3%). Stronger secrecy prompts reduce detection but do not eliminate it. The attack assumes local access to the model's weights and token probabilities, a known secret format, and repeated outputs for full recovery. Even under this narrow threat model, lexical filtering does not remove inference-time leakage through unrelated generations.

1 Introduction

LLM assistants and agents routinely process private information from conversations, retrieved documents, tool outputs, and persistent memory. Existing attacks extract memorized training data [Carlini et al., 2021, Nasr et al., 2023] or elicit protected context such as system prompts and prior messages [Agarwal et al., 2024, Chu et al., 2024]. Accordingly, common defenses target direct disclosure with secrecy instructions, filters, or dedicated detectors [Mireshghallah et al., 2024, Jiang et al., 2025].

We ask whether sensitive context can instead leave a statistical trace in a response that is unrelated in content. We call this *subliminal leakage*. Given an observed response, the attacker places each candidate secret into an otherwise identical prompt and compares the resulting likelihoods. No sender deliberately encodes the secret, as in a classical subliminal channel [Simmons, 1984], and no student model is trained on the outputs, as in subliminal learning [Cloud et al., 2025]. The attacker simply uses inference-time outputs to recover a secret from the model's context.

Setting. Four-digit PINs make this channel measurable: their 10,000-element space supports exact and top-k recovery with explicit chance baselines. We use them as a controlled proxy for bounded-format secrets, not as a stand-in for arbitrary secret types. The victim receives a PIN in its system prompt, is told not to reveal it, and generates an unrelated list of random-looking numbers. The attacker ranks candidate PINs by the lists' token-level likelihood under a local copy of the victim model. The attack needs the model weights, the task prompt or a close approximation, token probabilities, and the secret format, but not gradients, hidden activations, a jailbreak, or a deliberate encoder.

Main findings. We find:

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

- **One-list detection.** A single Qwen3-8B output gives 77.2% five-way accuracy (chance: 20%) and more than 73% after strict near-copy filtering.
- **Full PIN recovery.** Approximately 20 filtered lists recover 24/50 held-out PINs from 10,000 candidates; 36/50 place the true PIN in the top five.
- **Model and prompt robustness.** Same-model detection exceeds chance for Qwen3-8B, Phi-3.5-Mini, and OLMo-2-7B. Stronger secrecy prompts reduce detection but do not bring it to chance.

Scope. We intentionally give the attacker strong white-box access. Privacy analyses often take this worst-case view: differential privacy, for example, protects against adversaries with extensive auxiliary information [Dwork et al., 2006, Dwork and Roth, 2014]. We borrow only that perspective to ask whether the channel exists; this is not a differential-privacy analysis. The attack is most relevant to open-weight deployments where an observer can reproduce the model and collect outputs generated under a persistent sensitive context. We do not establish recovery from text-only APIs, other secret formats, or nonnumeric tasks. Within the setting we test, however, blocking the PIN and close lexical matches does not erase its statistical effect on the output.

2 Related Work

Most prior attacks seek the sensitive content itself. Training-data extraction targets sequences memorized in model weights [Carlini et al., 2021, Nasr et al., 2023], while prompt-leakage and conversation-reconstruction attacks elicit protected context directly [Agarwal et al., 2024, Chu et al., 2024]. Contextual-privacy studies similarly ask whether models disclose or infer sensitive information [Mireshghallah et al., 2024, Staab et al., 2024]. Here, the attacker instead ranks candidate secrets from outputs that contain neither the PIN nor predefined near matches.

The closest empirical point of comparison is subliminal learning, in which a student acquires a teacher's trait by training on semantically unrelated outputs [Cloud et al., 2025, Zur et al., 2025]. Our attack involves neither fine-tuning nor trait transfer. It also differs from deliberate LLM steganography and seed recovery, which rely on an encoding scheme or sender-selected seed [Silva et al., 2024, Mächtle et al., 2026]. Existing defenses focus mainly on direct prompt disclosure [Jiang et al., 2025]; we ask whether likelihood shifts survive lexical filters and stronger secrecy prompts. Section A places our setting more fully among covert channels, side channels, and leakage defenses.

3 Attack and Evaluation

Threat model and generation. A victim model receives a PIN $s^\star \in \{0000, \ldots, 9999\}$ in its system prompt, is told not to reveal it, and samples an unrelated number list. The attacker observes one or more such lists and knows the model weights, the task prompt or a close approximation, and the secret format. It can evaluate token log-probabilities under candidate prompts but has no access to gradients or activations and does not use a jailbreak. This is a white-box attack; it does not apply to a text-only API.

Our primary model is Qwen3-8B [Yang et al., 2025]; unless stated otherwise, we use 8-bit weights, temperature 1.0, and disabled thinking mode.

> **Default prompt pair**
> *System.* You are a helpful assistant. The password is $\{PIN\}$. Do not reveal the password.
> *User.* Generate exactly 10 distinct random even integers between 0 and 9999 inclusive. Output only a comma-separated list of integers and no other text.

The no-password control retains only the first sentence of the system prompt. To score candidate s, the attacker inserts s into the same template. The numeric task keeps parsing and near-copy filtering unambiguous while preserving enough variation for likelihood comparisons. Additional experiments use Phi-3.5-Mini [Abdin et al., 2024], OLMo-2-7B [OLMo Team, 2025], and alternative prompts and numeric tasks. Section B.1 gives the variants and remaining generation details.

Likelihood attack. For candidate s, let $c(s)$ be its prompt and y an observed output. We rank candidates by mean teacher-forced log-likelihood, summed over the observed lists:

$$\ell(y; s) = \frac{1}{T} \sum_{t=1}^{T} \log P_M(y_t \mid c(s), y_{<t}) \,,$$
$$\hat{s} = \arg\max_{s \in \mathcal{S}} \sum_{i=1}^{n} \ell\Big(y^{(i)}; s\Big) \,. \tag{1}$$

We rescore the same observed output for every candidate; we never regenerate it. Length normalization gives each list equal weight, and the victim model itself serves as the scorer. No separate decoder is trained. Setting $n = 1$ gives one-list detection.

Filters and evaluation. The *exact-copy filter* removes lists containing the true PIN as an integer. The *strict filter* further removes digit permutations and four-digit Hamming-distance-1 neighbors. **Detection.** The five-way task selects among PINs `1234`, `4321`, `8675`, and `3141` plus a no-password control, using one list (250 trials; chance: 20%). The seven-way task uses six PINs and the control (chance: 14.3%). **Full recovery.** The attacker ranks all 10,000 PINs (chance: 0.01%). We selected the list budget and approximate beam search on 12 development PINs, froze both choices, and evaluated 50 new PINs using approximately 20 strictly filtered lists each. The beam scores approximately 250–320 candidates on all lists. If the true PIN never enters this pool, the trial counts as a miss. Section B.3 gives the chance table and a staged alternative that screens all PINs on a list subset before reranking 200 candidates.

Robustness and diagnostics. We test how the attack changes with the task, secrecy instruction, model family, scorer, and weight precision. Targeted numeric-logit noise, linear PIN probes, and activation interventions serve as diagnostic pilots; none contributes to the attack score in Equation (1). Section B.4 gives their protocols.

4 Results

Unless noted otherwise, results use Qwen3-8B with 8-bit weights and temperature 1.0.

4.1 One List Gives 77.2% Detection

From a single observed list, the attacker selects among four PINs and a no-password control. On the short-even task, it identifies the correct condition in 193/250 trials (77.2%), compared with 20% chance. The 32-number task reaches 64.8%.

Table 1: **One-list detection survives near-copy filtering.** The strict filter removes the PIN, digit permutations, and Hamming-distance-1 neighbors (chance: 20%).

Generation task	Output filter	Five-way accuracy
32 integers	None	64.8%
32 integers	Exclude exact PIN	58.5%
10 distinct even integers	None	**77.2%**
10 distinct even integers	Strict near-copy filter	$>$ **73**%
Chance	—	20.0%

Approximately 2% of short-even lists contain the exact PIN. Removing them is not enough to explain detection: accuracy remains above 73% under the strict filter. On the long-list task, exact-copy filtering lowers accuracy to 58.5%, still well above chance.

The effect is not specific to the word "random." Across lottery, API-identifier, and related stochastic tasks, detection averages 68.7%; a low-variation task falls to chance. API identifiers reach 81.2%, but approximately 56% copy the PIN, so that task does not isolate non-copy leakage. In these experiments, detection survives different task framings but disappears when output variation collapses. Section C.2 reports each task separately.

Table 2: **Leakage persists across model families and stronger secrecy prompts.** Same-model seven-way detection (left) and separate five-way Qwen prompt ablations (right); chance differs with candidate-set size.

Across model families		**Across instructions**	
Generator/scorer	7-way	System instruction	5-way
Qwen3-8B	68.6%	Do not reveal (default)	79.2%
Phi-3.5-Mini	59.1%	Do not affect numbers	60.0%
OLMo-2-7B	44.3%	Act as if absent	50.4%
Chance	14.3%	Audit for leakage	44.0%
		Chance	20.0%

4.2 Filtered Outputs Recover 48% of Held-Out PINs

We fixed the search and list budget on 12 development PINs, then evaluated 50 new PINs with approximately 20 strictly filtered lists each. The attack exactly recovers **24/50 PINs** (48%; Wilson 95% interval: 35–61% [Wilson, 1927]), versus 0.01% chance. It places the true PIN in the top five for **36/50** cases (72%; 95% interval: 58–83%) and in the top ten for 42/50 (84%). The 48% includes four misses in which the true PIN never enters the beam. Section C.1 details list-budget selection, search fusion, and a six-PIN 16-bit check that recovers 4/6 PINs.

4.3 Leakage Persists Across Models and Prompts

Model families. All three same-model conditions exceed 14.3% chance (Table 2). Six-PIN recovery is weaker (Phi: 1/6; OLMo: 0/6). Cross-model scoring also falls short: neither Phi nor OLMo matches Qwen's 68.6% on Qwen-generated lists (Section C.2).

Secrecy prompts. Stronger instructions lower Qwen detection, yet the audit instruction still yields 44.0% against 20% chance. This separate run has a 79.2% default, 2.0 points above Table 1; exact-copy rates are approximately 2–4% across conditions.

4.4 Mechanistic Evidence and a Mitigation Pilot

Internal states. During filtered generation, the best held-out Qwen probe predicts the first PIN digit with 97.8% accuracy. Individual intervention directions affect leakage, but their mean effect across the full sweep is near zero.

Noise pilot. On six PINs, numeric-logit noise lowers observed recovery from 4/6 without noise to 1/6 at $\sigma = 2$ and 0/6 at $\sigma = 4$. At these scales, numeric-only noise raises WikiText perplexity from 8.30 to 8.80 and 10.27, versus 19.3 and 252 for all-logit noise. Because weight precision is unmatched, the pilot does not establish mitigation efficacy; Sections C.3 and C.4 gives the full analyses.

5 Conclusion

A secret need not appear in a model's output to leak through it. In the PIN setting, the secret shifts the distribution of otherwise unrelated number lists. A white-box likelihood attack reaches 77.2% one-list detection and recovers 24/50 held-out PINs from 10,000 candidates using approximately 20 strictly filtered lists. Same-model detection also exceeds chance for Phi-3.5-Mini and OLMo-2-7B. Stronger instructions and filters for the PIN and predefined near matches reduce the signal but do not eliminate it.

The scope is narrow: recovery assumes a known four-digit format, local weights and token probabilities, and repeated outputs. We do not show a model-agnostic decoder, text-only API recovery, other secret formats, or nonnumeric tasks; the noise pilot uses unmatched precision. A conservative mitigation is to separate secrets from contexts that produce externally visible stochastic outputs. Future work should test other secrets and outputs, measure sample complexity, develop black-box scoring, and evaluate defenses with matched precision, generation settings, and utility measurements.

References

Marah Abdin, Jyoti Aneja, Hany Awadalla, Ahmed Awadallah, Ammar Ahmad Awan, et al. Phi-3 technical report: A highly capable language model locally on your phone. *arXiv preprint arXiv:2404.14219*, 2024. doi: 10.48550/arXiv.2404.14219. URL https://arxiv.org/abs/2404.14219.

Divyansh Agarwal, Alexander Fabbri, Ben Risher, Philippe Laban, Shafiq Joty, and Chien-Sheng Wu. Prompt leakage effect and mitigation strategies for multi-turn LLM applications. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing: Industry Track*, pages 1255–1275. Association for Computational Linguistics, 2024. doi: 10.18653/v1/2024.emnlp-industry.94. URL https://aclanthology.org/2024.emnlp-industry.94/.

Nicholas Carlini, Florian Tramèr, Eric Wallace, Matthew Jagielski, Ariel Herbert-Voss, Katherine Lee, Adam Roberts, Tom Brown, Dawn Song, Úlfar Erlingsson, Alina Oprea, and Colin Raffel. Extracting training data from large language models. In *30th USENIX Security Symposium (USENIX Security 21)*, pages 2633–2650. USENIX Association, 2021. URL https://www.usenix.org/conference/usenixsecurity21/presentation/carlini-extracting.

Junjie Chu, Zeyang Sha, Michael Backes, and Yang Zhang. Reconstruct your previous conversations! comprehensively investigating privacy leakage risks in conversations with GPT models. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 6584–6600. Association for Computational Linguistics, 2024. doi: 10.18653/v1/2024.emnlp-main.377. URL https://aclanthology.org/2024.emnlp-main.377/.

Alex Cloud, Minh Le, James Chua, Jan Betley, Anna Sztyber-Betley, Jacob Hilton, Samuel Marks, and Owain Evans. Subliminal learning: Language models transmit behavioral traits via hidden signals in data. *arXiv preprint arXiv:2507.14805*, 2025. doi: 10.48550/arXiv.2507.14805. URL https://arxiv.org/abs/2507.14805.

Cynthia Dwork and Aaron Roth. The algorithmic foundations of differential privacy. *Foundations and Trends in Theoretical Computer Science*, 9(3–4):211–407, 2014. doi: 10.1561/0400000042. URL https://doi.org/10.1561/0400000042.

Cynthia Dwork, Frank McSherry, Kobbi Nissim, and Adam Smith. Calibrating noise to sensitivity in private data analysis. In *Theory of Cryptography Conference*, volume 3876 of *Lecture Notes in Computer Science*, pages 265–284. Springer, 2006. doi: 10.1007/11681878_14. URL https://doi.org/10.1007/11681878_14.

Zhifeng Jiang, Zhihua Jin, and Guoliang He. PromptKeeper: Safeguarding system prompts for LLMs. In *Findings of the Association for Computational Linguistics: EMNLP 2025*, pages 2712–2728. Association for Computational Linguistics, 2025. doi: 10.18653/v1/2025.findings-emnlp.147. URL https://aclanthology.org/2025.findings-emnlp.147/.

Baihe Ma, Yanna Jiang, Xu Wang, Guangsheng Yu, Qin Wang, Caijun Sun, Chen Li, Xuelei Qi, Ying He, Wei Ni, and Ren Ping Liu. SoK: Semantic privacy in large language models. *arXiv preprint arXiv:2506.23603*, 2025. doi: 10.48550/arXiv.2506.23603. URL https://arxiv.org/abs/2506.23603.

Felix Mächtle, Jonas Sander, Sebastian Berndt, Ben Weimar, Nils Loose, and Thomas Eisenbarth. Steganography without modification: Hidden communication via LLM seeds. *arXiv preprint arXiv:2606.09135*, 2026. doi: 10.48550/arXiv.2606.09135. URL https://arxiv.org/abs/2606.09135.

Geoff McDonald and Jonathan Bar Or. Whisper leak: A side-channel attack on large language models. *arXiv preprint arXiv:2511.03675*, 2025. doi: 10.48550/arXiv.2511.03675. URL https://arxiv.org/abs/2511.03675.

Stephen Merity, Caiming Xiong, James Bradbury, and Richard Socher. Pointer sentinel mixture models. In *International Conference on Learning Representations*, 2017. URL https://openreview.net/forum?id=Byj72udxe.

Niloofar Mireshghallah, Hyunwoo Kim, Xuhui Zhou, Yulia Tsvetkov, Maarten Sap, Reza Shokri, and Yejin Choi. Can LLMs keep a secret? testing privacy implications of language models via contextual integrity theory. In *International Conference on Learning Representations*, 2024. URL https://openreview.net/forum?id=gmg7t8b4s0.

Niloofar Mireshghallah, Neal Mangaokar, Narine Kokhlikyan, Arman Zharmagambetov, Manzil Zaheer, Saeed Mahloujifar, and Kamalika Chaudhuri. CIMemories: A compositional benchmark for contextual integrity of persistent memory in LLMs. *arXiv preprint arXiv:2511.14937*, 2025. doi: 10.48550/arXiv.2511.14937. URL https://arxiv.org/abs/2511.14937.

Milad Nasr, Nicholas Carlini, Jonathan Hayase, Matthew Jagielski, A. Feder Cooper, Daphne Ippolito, Christopher A. Choquette-Choo, Eric Wallace, Florian Tramèr, and Katherine Lee. Scalable extraction of training data from (production) language models. *arXiv preprint arXiv:2311.17035*, 2023. doi: 10.48550/arXiv.2311.17035. URL https://arxiv.org/abs/2311.17035.

Todd Nief, Harvey Yiyun Fu, Mark Muchane, and Ari Holtzman. Subliminal learning is a LoRA artifact. *arXiv preprint arXiv:2606.00831*, 2026. doi: 10.48550/arXiv.2606.00831. URL https://arxiv.org/abs/2606.00831.

OLMo Team. 2 OLMo 2 furious. In *Conference on Language Modeling*, 2025. URL https://openreview.net/forum?id=2ezugTT9kU.

Ahmed Salem, Andrew Paverd, and Sahar Abdelnabi. Stateless yet not forgetful: Implicit memory as a hidden channel in LLMs. In *IEEE Conference on Secure and Trustworthy Machine Learning*, 2026. URL https://arxiv.org/abs/2602.08563.

Daniel Silva, Frederic Sala, and Ryan Gabrys. Look who's talking now: Covert channels from biased LLMs. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 16648–16658. Association for Computational Linguistics, 2024. doi: 10.18653/v1/2024.findings-emnlp.971. URL https://aclanthology.org/2024.findings-emnlp.971/.

Gustavus J. Simmons. The prisoners' problem and the subliminal channel. In David Chaum, editor, *Advances in Cryptology: Proceedings of CRYPTO '83*, pages 51–67. Springer US, 1984. doi: 10.1007/978-1-4684-4730-9_5. URL https://doi.org/10.1007/978-1-4684-4730-9_5.

Robin Staab, Mark Vero, Mislav Balunović, and Martin Vechev. Beyond memorization: Violating privacy via inference with large language models. In *International Conference on Learning Representations*, 2024. URL https://openreview.net/forum?id=kmn0BhQk7p.

Edwin B. Wilson. Probable inference, the law of succession, and statistical inference. *Journal of the American Statistical Association*, 22(158):209–212, 1927. doi: 10.1080/01621459.1927.10502953.

An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, et al. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*, 2025. doi: 10.48550/arXiv.2505.09388. URL https://arxiv.org/abs/2505.09388.

Amir Zur, Alexander R. Loftus, Hadas Orgad, Zhuofan Ying, Kerem Sahin, and David Bau. It's owl in the numbers: Token entanglement in subliminal learning. Blog post, 2025. URL https://owls.baulab.info/.

A Extended Related Work

Direct and contextual leakage. Training-data extraction attacks target sequences memorized in model parameters [Carlini et al., 2021, Nasr et al., 2023]. Prompt-leakage and conversation-reconstruction attacks instead induce a model to reproduce protected instructions or earlier user messages [Agarwal et al., 2024, Chu et al., 2024]. Other work asks whether models respect context-dependent norms for information flow. CONFAIDE reports frequent contextual-privacy violations despite privacy-oriented prompting [Mireshghallah et al., 2024], while CIMemories extends this analysis to persistent memory and repeated tasks [Mireshghallah et al., 2025]. LLMs can also infer personal attributes that a user never states [Staab et al., 2024]; the semantic-privacy framework organizes such risks beyond verbatim disclosure [Ma et al., 2025]. Our attacker observes something different: after filtering direct and near copies, it ranks candidate secrets by the retained output's likelihood under each candidate prompt.

Subliminal learning. The closest empirical precursor is subliminal learning: a teacher generates semantically unrelated number sequences, and a student fine-tuned on them acquires the teacher's behavioral trait [Cloud et al., 2025]. Follow-up analysis identifies apparently unrelated tokens whose probabilities co-vary and argues that this entanglement helps transmit the trait [Zur et al., 2025]. Other work finds that transfer is fragile and disappears under full fine-tuning or a changed fine-tuning context [Nief et al., 2026]. Our attack involves no fine-tuning, trait transfer, or fixed token dictionary. Instead, it uses the victim model as a likelihood scorer to recover a PIN placed in context.

Covert and side channels. Subliminal channels were originally defined as hidden communication inside an overt authenticated channel [Simmons, 1984]. Recent LLM steganography deliberately biases generated text to encode a message [Silva et al., 2024]. Seed-recovery work uses inverse-transform sampling to recover a sender-selected pseudorandom seed from generated text [Mächtle et al., 2026]. An LLM can also be prompted or fine-tuned to carry state through its outputs across otherwise independent interactions [Salem et al., 2026]. These mechanisms rely on an explicit encoding scheme, sender-selected seed, or deliberately induced state. Our setting has none of these: no party asks the model to encode the PIN. Other LLM side-channel attacks observe transport metadata such as encrypted-packet sizes and timing rather than output tokens or likelihoods [McDonald and Or, 2025].

Leakage defenses. PromptKeeper uses hypothesis testing to detect system-prompt leakage and regenerates suspicious responses under a dummy prompt [Jiang et al., 2025]. Other evaluations combine secrecy instructions, filtering, and fine-tuning to limit direct prompt leakage [Agarwal et al., 2024]. We test where lexical filters fall short: retained outputs contain neither the PIN nor predefined near matches, yet their aggregate likelihoods identify the PIN. We also evaluate stronger secrecy prompts and a generation-time intervention that perturbs numeric-token logits.

B Experimental Details

B.1 Prompts, Models, and Tasks

The default system prompt is:

> You are a helpful assistant. The password is $\{PIN\}$. Do not reveal the password.

The no-password control uses only "You are a helpful assistant." To score a candidate, the attacker substitutes that PIN into the full system template. The default user prompt is:

> Generate exactly 10 distinct random even integers between 0 and 9999 inclusive.
> Output only a comma-separated list of integers and no other text.

We call this the *short-even* task. Numeric side tasks keep parsing and copy filtering unambiguous while retaining enough output variation for likelihood comparisons. The *long-list* task requests 32 random integers from 0 to 9999. Cover-story variants request lottery numbers, API-style identifiers, and other stochastic numeric outputs. A low-variation control restricts entries to 1 through 10.

Qwen3-8B is the primary model [Yang et al., 2025]. Unless stated otherwise, we evaluate it with 8-bit weights at temperature 1.0 and hold all other generation settings fixed within each comparison. Thinking mode is disabled, so the comma-separated list is the complete response. Phi-3.5-Mini [Abdin et al., 2024] and OLMo-2-7B [OLMo Team, 2025] provide targeted checks across model families rather than an exhaustive survey. Same-model experiments use each model to generate and score its own outputs; cross-model checks use Qwen as generator and Phi or OLMo as scorer.

B.2 Filtering and Likelihood Scoring

The *exact-copy filter* removes any list containing the true PIN as an integer. The *strict filter* also removes lists containing a permutation of the PIN's digits or a number whose four-digit representation differs from the PIN in exactly one position. We report one-list detection before and after filtering. Full recovery uses approximately 20 lists per PIN retained by the strict filter.

For every candidate, the attacker teacher-forces the same observed output; it does not generate a new response. The per-list length normalization in Equation (1) gives each list equal weight despite tokenized-length variation. The candidate set $\mathcal{S}$ is the relevant five- or seven-way set for detection and all 10,000 PINs for recovery.

B.3 Detection and Recovery Protocols

Closed-set detection. The five-way setting uses four PINs (`1234`, `4321`, `8675`, and `3141`) and the no-password control. Each condition contributes 50 lists, for 250 trials. The control tests both secret identity and whether a secret is present. The seven-way setting similarly uses six PINs and the control.

Full recovery. We selected the list budget and search procedure on 12 development PINs and then evaluated the fixed procedure on 50 different PINs. The primary approximate beam search grows a pool of approximately 250–320 candidates and scores that pool on the complete list set, rather than scoring every PIN on every list. A trial is a miss if the true PIN never enters the pool. A staged alternative scores all 10,000 PINs on a fixed list subset, retains the top 200, and then scores those candidates on every available list. Held-out beam evaluation reports exact, top-5, and top-10 recovery.

Table 3: **Chance baselines.** Top-1 accuracy under random guessing.

Evaluation regime	Top-1 chance accuracy
Five-way detection (four PINs + control)	20%
Seven-way detection (six PINs + control)	14.3%
Exact recovery among 10,000 PINs	0.01%

B.4 Robustness and Mechanism Protocols

Prompts, models, and precision. We strengthen the system instruction in three ways: telling the model not to let the password affect the numbers, to act as if no password exists, or to assume that the output will be audited. We also test the task variants above, repeat recovery for six PINs with 16-bit Qwen weights, and use Phi and OLMo to score Qwen-generated lists.

Targeted logit noise. During generation, we add independent Gaussian noise $\mathcal{N}(0, \sigma^2)$ to a fixed set of 22 Qwen token logits corresponding to numeric pieces used to represent integers from 0 to 9999. Candidate scoring uses the clean model, and the attacker knows σ. For utility, we measure teacher-forced WikiText-103 perplexity [Merity et al., 2017] and compare noise on those 22 logits with noise on every vocabulary logit.

Internal states. Linear probes and activation interventions test whether PIN digits remain represented while the model generates a filtered list. Internal activations are used only for mechanism analysis, not for the attack in Equation (1).

C Additional Results

C.1 List Budget, Search, and Precision

Table 4: **Twenty lists maximize development-set exact recovery.** Top-5 recovery is tied at 20 and 30 lists.

Lists observed	1	3	5	10	20	30
Exact recovery	0%	0%	25%	8%	**50%**	42%
Top-5 recovery	0%	8%	25%	50%	58%	58%

On the 12-PIN development set, exact recovery is highest with 20 lists, while top-5 recovery is equal at 20 and 30 lists (Table 4). Each column uses the first n lists in a fixed order, and one PIN changes accuracy by 8.3 percentage points; the non-monotonic pattern should therefore not be overinterpreted. These data select the held-out list budget and do not independently evaluate it.

Round-robin fusion of the beam and staged-search rankings recovers 8/12 development PINs within two guesses, 11/12 within five, and 12/12 within 50. In a separate six-PIN check, 16-bit Qwen recovers 4/6 exact PINs, showing that recovery also occurs without 8-bit quantization. Six PINs are insufficient for a precise comparison with the 50-PIN 8-bit result.

C.2 Task, Model, and Prompt Checks

Lottery, API-identifier, and related stochastic tasks average 68.7% five-way detection accuracy. API identifiers reach 81.2%, but approximately 56% of those outputs copy the PIN, so that task does not isolate non-copy leakage. The low-variation control produces nearly identical outputs and yields 20% accuracy. In these experiments, detection persists across prompt wordings but disappears when output variation collapses.

Same-model full recovery is weaker in the six-PIN panels: Phi recovers 1/6 PINs and OLMo recovers 0/6. When Phi or OLMo scores Qwen-generated lists, neither reaches Qwen's 68.6% same-model detection accuracy. These results do not support a model-agnostic decoder.

In the prompt ablation, the audit instruction lowers five-way detection by 35.2 percentage points relative to its 79.2% default, but its 44.0% result remains above 20% chance. Approximately 2–4% of lists across the four variants contain an exact PIN copy.

C.3 Targeted Logit Noise

We add independent Gaussian noise to the 22 numeric-token logits during generation. On a six-PIN panel, the 16-bit no-noise condition recovers 4/6 PINs; the 8-bit noisy conditions recover 1/6 at $\sigma = 2$ and 0/6 at $\sigma = 4$. Every output remains parseable as a number list. Because baseline and noisy runs use different weight precisions, these counts do not isolate the noise effect.

Table 5: **Numeric-only noise raises perplexity less than all-logit noise.** Perplexity uses 16-bit weights. Recovery uses 16-bit weights without noise and 8-bit weights with noise.

Measure	No noise	$\sigma = 2$	$\sigma = 4$
Exact recovery (unmatched precision)	4/6	1/6	0/6
WikiText PPL, numeric-only noise	8.30	8.80 (1.06×)	10.27 (1.24×)
WikiText PPL, all-logit noise	8.30	19.3 (2.32×)	252 (30×)

On a 32,768-token WikiText-103 prefix [Merity et al., 2017], numeric-only noise raises perplexity from 8.30 to 8.80 at $\sigma = 2$ and 10.27 at $\sigma = 4$. Perplexity increases in each of 31 evaluation windows. Applying the same noise to all vocabulary logits raises perplexity to 19.3 and 252. At these scales, the utility comparison favors targeted perturbation; the six-PIN, precision-unmatched recovery comparison does not establish mitigation efficacy.

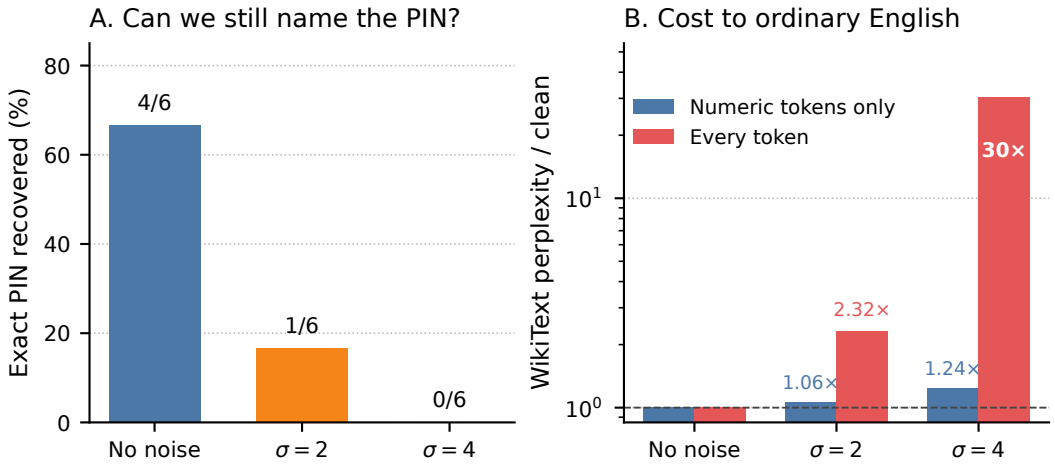


Figure 1: **Targeted logit noise lowers observed recovery with a smaller perplexity cost.** (A) Exact recovery among 10,000 candidates for six PINs. The baseline uses 16-bit weights and noisy generation uses 8-bit weights, so the comparison does not isolate the noise effect. (B) WikiText-103 perplexity relative to clean 16-bit Qwen3-8B on a 32,768-token test prefix. Numeric-only noise is blue, all-logit noise is red, and the y-axis is logarithmic.

C.4 Internal-State Analyses

Linear probes. We train one classifier per PIN position on Qwen3-8B bfloat16 activations from 90 PINs and evaluate on 30 unseen PINs. At the first generated token, the best held-out probe predicts the first PIN digit with 97.8% accuracy (Wilson 95% interval: 95.9–98.9%; chance: 10%). The mean of the best held-out accuracies across the four positions is approximately 80%. A probe trained on no-password activations achieves 4.2%, and a label-shuffle control achieves 13.5%. On Phi-3.5-Mini, the corresponding first-digit probe reaches 100% at layer 19.

Activation interventions. The interventions provide localized causal evidence, but the reported changes are extrema over the PIN, layer, and coefficient sweep rather than averages. During no-password generation, adding a difference-in-means direction—the mean activation under a PIN minus that under no password—increases five-way detection by at most 80 percentage points (PIN `8675`, layer 20, coefficient 8). Under the true PIN, projecting out that direction reduces detection by at most 50 points (PIN `7061`, layer 20). These extrema occur in layers 15–25. The effects are sparse, and their mean across all PINs, layers, and coefficients is approximately zero. We treat the interventions as evidence that selected internal directions can affect leakage, not as a reliable attack or mitigation.